

02 - Technical Operator Runbook / ขั้นตอนปฏิบัติทางเทคนิคสำหรับ Admin

ScheduledDefrag Trigger Validation + Cross-Layer Hang/Delay RCA

Windows Server 2022 + Veritas + VMware + HPE Alletra 9060

Important Concept / แนวคิดสำคัญ

English

Steps 4-16 are the **standard execution block for one scenario**.

You must repeat Steps 4-16 for each scenario:

```
Scenario A: C:\Temp\TrimTest-A-Baseline  
Scenario B: C:\Temp\TrimTest-B-DefragDisabled  
Scenario C: C:\Temp\TrimTest-C-Mitigation
```

Each scenario must use a separate OutputDir.

Each scenario requires a large SQL backup file to exist before deletion.

If the same file is used, recreate or copy the file again before starting the next scenario.

Main-plan note:

```
Scenario A/B/C do not include a manual Optimize-Volume / ReTrim command.  
If manual optimizer confirmation is needed, run it only as an optional  
appendix test after the main A/B/C plan is complete.
```

ภาษาไทย

Step 4-16 คือ **standard execution block** สำหรับ 1 scenario

ต้องทำซ้ำ Step 4-16 สำหรับแต่ละ scenario:

```
Scenario A: C:\Temp\TrimTest-A-Baseline  
Scenario B: C:\Temp\TrimTest-B-DefragDisabled
```

Scenario C: C:\Temp\TrimTest-C-Mitigation

แต่ละ scenario ต้องใช้ OutputDir แยกกัน เพื่อไม่ให้ evidence ปนกัน

แต่ละ scenario ต้องมีไฟล์ SQL backup ขนาดใหญ่ให้ลบก่อนเริ่มทดสอบ
ถ้าใช้ไฟล์เดิม ต้อง copy/recreate ไฟล์กลับมาก่อนเริ่ม scenario ถัดไป

Common Preparation / เตรียมก่อนเริ่มทดสอบ

Step 1 - Prepare Script Folder / เตรียม Folder สำหรับ Script

Where:

Windows Server 2022 PowerShell – Run as Administrator

Estimated:

5 minutes

Run:

```
mkdir C:\Temp\TrimTest-Scripts  
cd C:\Temp\TrimTest-Scripts
```

Copy these scripts into C:\Temp\TrimTest-Scripts:

```
03-TrimLatencyTrace-kevin.ps1  
04-Start-PerfMonBLG.ps1  
05-Stop-PerfMonBLG.ps1  
06-Collect-StorageEventSnapshot.ps1  
07-Collect-VeritasWindowsEvidence.ps1  
08-Export-WindowsTrimEvidence.ps1
```

Expected:

Scripts are available in C:\Temp\TrimTest-Scripts

Kevin version note:

```
Use 03-TrimLatencyTrace-kevin.ps1 instead of 03-TrimLatencyTrace.ps1.  
This Kevin version automatically captures:  
- ScheduledDefrag task configuration by schtasks /query  
- Win32_Process details every 60 seconds for suspected optimizer activity
```

Step 2 - Prepare External Collector Workstations / เตรียมเครื่องมือภายนอก

Where:

```
VMware PowerCLI workstation  
HPE Alletra 9060 management workstation
```

Estimated:

```
10 minutes
```

Run / Prepare:

```
VMware workstation:  
09-Collect-VMwareLatency.ps1  
  
HPE Alletra 9060 management workstation:  
10-Collect-3PARLatency.ps1
```

Expected:

```
VMware PowerCLI access confirmed  
HPE Alletra 9060 SSH access confirmed
```

Step 3 - Prepare Customer Test File Prerequisites / เตรียมความพร้อมของไฟล์ทดสอบจากฝั่งลูกค้า

Where:

```
Customer application / backup owner  
Windows Server 2022
```

Estimated:

This should be completed before the test window starts.

Confirm:

Item	What must be confirmed
File path	Full path of the large SQL backup file
File size	Target size, ideally around 1.2 TB or agreed equivalent
File type	Example: .bak
Target volume	Drive letter / volume that will be tested
Delete approval	Customer confirms the file can be safely deleted
Recreate method	Copy back / restore / regenerate method for next scenario
Recreate owner	Person/team responsible for preparing the file again
Recreate duration	Approximate time needed before the next scenario

Expected:

The customer-side file preparation plan is confirmed before execution day.

Step 4 - Confirm Target File and Recreate Plan / ยืนยันไฟล์ที่จะลบ

Where:

Windows Server 2022

Estimated:

5 minutes

Run / Confirm:

Confirm large SQL backup file path.
Confirm file can be safely deleted.
Confirm how the file will be recreated/copied for next scenario.

Example file path:

```
E:\Backup\LargeSqlBackup.bak
```

Expected:

```
Target file path confirmed.
File recreation plan confirmed.
```

Scenario Configuration Matrix / ตารางตั้งค่า Scenario

Use the configuration below before running the standard execution block.

Scenario	OutputDir	DisableDeleteNotify	ScheduledDefrag
A - Baseline	C:\Temp\TrimTest-A-Baseline	0	Enabled/current
B - Defrag Disabled	C:\Temp\TrimTest-B-DefragDisabled	0	Disabled
C - Mitigation	C:\Temp\TrimTest-C-Mitigation	1	Disabled

Scenario A - Baseline / Reproduce

Configure Scenario A

Where:

```
Windows Server 2022 PowerShell – Run as Administrator
```

Estimated:

```
5 minutes
```

Run:

```
$OutputDir = "C:\Temp\TrimTest-A-Baseline"

fsutil behavior set DisableDeleteNotify 0
fsutil behavior query DisableDeleteNotify

Enable-ScheduledTask -TaskPath "\Microsoft\Windows\Defrag\" -TaskName
```

```
"ScheduledDefrag"
```

```
Get-ScheduledTask -TaskPath "\Microsoft\Windows\Defrag\" -TaskName  
"ScheduledDefrag" |  
Select-Object TaskName, State  
  
Get-Process defrag -ErrorAction SilentlyContinue
```

Expected:

```
NTFS DisableDeleteNotify = 0  
ScheduledDefrag enabled/current  
No unexpected defrag.exe before start
```

Then run:

```
Step 5-17 using OutputDir = C:\Temp\TrimTest-A-Baseline
```

Scenario B - Trigger Isolation / ScheduledDefrag Disabled

Configure Scenario B

Where:

```
Windows Server 2022 PowerShell – Run as Administrator
```

Estimated:

```
5 minutes
```

Run:

```
$OutputDir = "C:\Temp\TrimTest-B-DefragDisabled"  
  
fsutil behavior set DisableDeleteNotify 0  
fsutil behavior query DisableDeleteNotify  
  
Disable-ScheduledTask -TaskPath "\Microsoft\Windows\Defrag\" -TaskName  
"ScheduledDefrag"
```

```
Get-ScheduledTask -TaskPath "\Microsoft\Windows\Defrag\" -TaskName  
"ScheduledDefrag" |  
Select-Object TaskName, State  
  
Get-Process defrag -ErrorAction SilentlyContinue
```

Expected:

```
NTFS DisableDeleteNotify = 0  
ScheduledDefrag disabled  
No defrag.exe running
```

Then run:

```
Step 5-17 using OutputDir = C:\Temp\TrimTest-B-DefragDisabled
```

Scenario C - Mitigation Validation / Known-Good Control

Configure Scenario C

Where:

```
Windows Server 2022 PowerShell – Run as Administrator
```

Estimated:

```
5 minutes
```

Run:

```
$OutputDir = "C:\Temp\TrimTest-C-Mitigation"  
  
fsutil behavior set DisableDeleteNotify 1  
fsutil behavior query DisableDeleteNotify  
  
Disable-ScheduledTask -TaskPath "\Microsoft\Windows\Defrag\" -TaskName  
"ScheduledDefrag"  
  
Get-ScheduledTask -TaskPath "\Microsoft\Windows\Defrag\" -TaskName
```

```
"ScheduledDefrag" |  
Select-Object TaskName, State  
  
Get-Process defrag -ErrorAction SilentlyContinue
```

Expected:

```
NTFS DisableDeleteNotify = 1  
ScheduledDefrag disabled  
No defrag.exe running
```

Then run:

```
Step 5-17 using OutputDir = C:\Temp\TrimTest-C-Mitigation
```

Standard Execution Block / ขั้นตอนมาตรฐานสำหรับ 1 Scenario

Step 5 - Create Scenario Output Folder / สร้าง Output Folder

Where:

```
Windows Server 2022 PowerShell – Run as Administrator
```

Estimated:

```
1 minute
```

Run:

```
New-Item -ItemType Directory -Path $OutputDir -Force | Out-Null
```

Expected:

```
Scenario OutputDir created.
```

Step 6 - Collect Pre-check Storage Snapshot / เก็บ Storage Snapshot ก่อนทดสอบ

Where:

```
Windows Server 2022 PowerShell – Run as Administrator
```

Estimated:

```
5 minutes
```

Run:

```
cd C:\Temp\TrimTest-Scripts  
.\06-Collect-StorageEventSnapshot.ps1 -OutputDir "$OutputDir\PreCheck"
```

Expected:

```
storage-snapshot-*.zip created under <OutputDir>\PreCheck
```

Step 7 - Collect Baseline Veritas Evidence / เก็บ Veritas Baseline

Where:

```
Windows Server 2022 PowerShell – Run as Administrator
```

Estimated:

```
10 minutes
```

Run:

```
cd C:\Temp\TrimTest-Scripts  
.\07-Collect-VeritasWindowsEvidence.ps1 -OutputDir "$OutputDir\Veritas"
```

Expected:

```
veritas-windows-*.zip created under <OutputDir>\Veritas
```

Step 8 - Start Windows Collectors / เริ่ม Windows Collectors

Where:

```
Windows Server 2022 PowerShell – Run as Administrator
```

Estimated:

```
5 minutes
```

Run:

```
cd C:\Temp\TrimTest-Scripts

.\03-TrimLatencyTrace-kevin.ps1 -Action Start -OutputDir $OutputDir -
SampleInterval 5

.\04-Start-PerfMonBLG.ps1 -OutputDir $OutputDir -SampleInterval 5
```

Verify:

```
Get-Job
```

Expected:

```
TrimTest-Counters           Running
TrimTest-DefragProcess      Running
TrimTest-DefragEvents       Running
TrimTest-Fsutil             Running
TrimTest-KevinProcessCim    Running

disk_latency_perfmon.blg created under <OutputDir>
kevin-scheduleddefrag-task-config.txt created under <OutputDir>
kevin-process-cim-monitor.csv created under <OutputDir>
```

Step 9 - Start VMware Collector / เริ่ม VMware Collector

Where:

VMware PowerCLI workstation

Estimated:

5 minutes

Run:

```
.\09-Collect-VMwareLatency.ps1 `
  -vCenter "vcenter.example.com" `
  -VMName "SQL-Windows-VM" `
  -DatastoreName "DATASTORE_NAME" `
  -OutputDir "<ScenarioOutputDir>\VMware" `
  -SampleInterval 20
```

Example:

```
.\09-Collect-VMwareLatency.ps1 `
  -vCenter "vcenter.example.com" `
  -VMName "SQL-Windows-VM" `
  -DatastoreName "DATASTORE_NAME" `
  -OutputDir "C:\Temp\TrimTest-B-DefragDisabled\VMware" `
  -SampleInterval 20
```

Expected:

```
vmware-vm-disk-latency.csv
vmware-datastore-latency.csv
```

Important:

Keep this collector running until Step 15.

Step 10 - Start HPE Alletra 9060 Collector / เริ่ม HPE Alletra 9060 Collector

Where:

Management workstation with SSH access to HPE Alletra 9060

Estimated:

5 minutes

Run:

```
.\10-Collect-3PARLatency.ps1 `
  -Array "3par-array.example.com" `
  -User "monitoruser" `
  -HostFilter "SQL_HOST_NAME" `
  -OutputDir "<ScenarioOutputDir>\3PAR" `
  -SampleInterval 5
```

Example:

```
.\10-Collect-3PARLatency.ps1 `
  -Array "3par-array.example.com" `
  -User "monitoruser" `
  -HostFilter "SQL_HOST_NAME" `
  -OutputDir "C:\Temp\TrimTest-B-DefragDisabled\3PAR" `
  -SampleInterval 5
```

Expected:

```
3par-latency-*.log
3par-commands-*.txt
```

Important:

Keep this collector running until Step 15.

Step 11 - Mark Before Delete / Mark ก่อนลบไฟล์

Where:

Windows Server 2022 PowerShell – Run as Administrator

Estimated:

1 minute

Run:

```
cd C:\Temp\TrimTest-Scripts  
  
.\03-TrimLatencyTrace-kevin.ps1 -Action Mark -OutputDir $OutputDir -Message  
"Before delete SQL backup"
```

Expected:

```
timeline_markers.csv updated.
```

Step 12 - Delete Large SQL Backup File / ลบไฟล์ SQL Backup ขนาดใหญ่

Where:

Windows Server 2022 PowerShell – Run as Administrator

Estimated:

1-5 minutes

Run:

```
Remove-Item "E:\Backup\LargeSqlBackup.bak" -Force
```

Expected:

```
Large SQL backup file deleted.
```

Important:

Replace E:\Backup\LargeSqlBackup.bak with the actual file path.
Before running the next scenario, recreate/copy this file again.

Step 13 - Mark After Delete / Mark หลังลบไฟล์

Where:

Windows Server 2022 PowerShell – Run as Administrator

Estimated:

1 minute

Run:

```
cd C:\Temp\TrimTest-Scripts  
  
.\03-TrimLatencyTrace-kevin.ps1 -Action Mark -OutputDir $OutputDir -Message  
"After delete SQL backup"
```

Expected:

timeline_markers.csv updated.

Step 14 - Observe 60+ Minutes / Monitor อย่างน้อย 60 นาที

Where:

Windows Server 2022
VMware PowerCLI workstation
HPE Alletra 9060 management workstation

Estimated:

60 minutes minimum

Optional Windows status check:

```
cd C:\Temp\TrimTest-Scripts

.\03-TrimLatencyTrace-kevin.ps1 -Action Status -OutputDir $OutputDir
```

Watch for:

```
hang/slowness
defrag.exe activity
Defrag Operational events
Windows disk latency spike
Windows disk queue spike
VMware datastore latency spike
HPE Alletra 9060 latency spike
Veritas / path / disk events
```

If hang/slowness occurs:

```
cd C:\Temp\TrimTest-Scripts

.\06-Collect-StorageEventSnapshot.ps1 -OutputDir "$OutputDir\DuringSpike"

.\07-Collect-VeritasWindowsEvidence.ps1 -OutputDir "$OutputDir\Veritas"
```

Expected:

```
Observation completed with or without issue.
If issue occurs, DuringSpike evidence captured.
```

Step 15 - Stop Windows Collectors / หยุด Windows Collectors

Where:

Windows Server 2022 PowerShell – Run as Administrator

Estimated:

5 minutes

Run:

```
cd C:\Temp\TrimTest-Scripts  
.\03-TrimLatencyTrace-kevin.ps1 -Action Stop -OutputDir $OutputDir  
.\05-Stop-PerfMonBLG.ps1
```

Expected:

```
Windows collector jobs stopped.  
PerfMon BLG stopped.
```

Step 16 - Stop VMware and HPE Alletra 9060 Collectors / หยุด VMware และ HPE Alletra 9060

Where:

```
VMware PowerCLI workstation  
HPE Alletra 9060 management workstation
```

Estimated:

1 minute

Run:

```
Press Ctrl+C in each collector window.
```

Expected:

```
VMware CSV collection stopped.  
HPE Alletra 9060 log collection stopped.
```


Step 17 - Export Final Evidence / Export Evidence หลังจบ Scenario

Where:

```
Windows Server 2022 PowerShell – Run as Administrator
```

Estimated:

```
10–15 minutes
```

Run:

```
cd C:\Temp\TrimTest-Scripts

.\08-Export-WindowsTrimEvidence.ps1 -OutputDir $OutputDir

.\06-Collect-StorageEventSnapshot.ps1 -OutputDir "$OutputDir\PostTest"

.\07-Collect-VeritasWindowsEvidence.ps1 -OutputDir "$OutputDir\Veritas"
```

Expected:

```
windows-eventlogs-*.zip created
storage-snapshot-*.zip created
veritas-windows-*.zip created
```

After Each Scenario / หลังจบแต่ละ Scenario

Before moving to the next scenario:

1. Confirm all collectors are stopped.
2. Confirm evidence files exist in the scenario OutputDir.
3. Recreate/copy the large SQL backup file again.
4. Configure the next scenario.
5. Repeat Step 5–17.

Final Evidence Checklist / Checklist หลักฐานสุดท้าย

After all scenarios, evidence should be separated as:

```
C:\Temp\TrimTest-A-Baseline  
C:\Temp\TrimTest-B-DefragDisabled  
C:\Temp\TrimTest-C-Mitigation
```

Each folder should contain:

```
disk_latency_counters.csv  
disk_latency_perfmon.blg  
defrag_process_monitor.csv  
defrag_event_monitor.csv  
kevin-scheduleddefrag-task-config.txt  
kevin-process-cim-monitor.csv  
fsutil_trim_status.csv  
timeline_markers.csv  
environment_snapshot.txt  
summary_after_stop.txt  
windows-eventlogs-*.zip  
PreCheck\storage-snapshot-*.zip  
PostTest\storage-snapshot-*.zip  
Veritas\veritas-windows-*.zip  
VMware\vmware-vm-disk-latency.csv  
VMware\vmware-datastore-latency.csv  
3PAR\3par-latency-*.log  
3PAR\3par-commands-*.txt
```

Appendix A - Optional Manual Optimize-Volume / ReTrim Confirmation

Use this appendix only if the customer requests extra confirmation after Scenario A/B/C are complete.

Purpose:

```
Confirm whether a manually triggered optimizer/retrim pass reproduces the  
same delay pattern  
after the file deletion has already completed.
```

Important:

```
This appendix is optional and is not part of the main customer A/B/C plan.  
Use a separate OutputDir so the optional evidence does not mix with the
```

```
main evidence set.
```

Recommended OutputDir:

```
C:\Temp\TrimTest-D-ManualRetrim
```

Appendix A Step 1 - Configure Optional Scenario

Where:

```
Windows Server 2022 PowerShell – Run as Administrator
```

Run:

```
$OutputDir = "C:\Temp\TrimTest-D-ManualRetrim"

fsutil behavior set DisableDeleteNotify 0
fsutil behavior query DisableDeleteNotify

Disable-ScheduledTask -TaskPath "\Microsoft\Windows\Defrag\" -TaskName
"ScheduledDefrag"

Get-ScheduledTask -TaskPath "\Microsoft\Windows\Defrag\" -TaskName
"ScheduledDefrag" |
Select-Object TaskName, State

Get-Process defrag -ErrorAction SilentlyContinue
```

Expected:

```
DisableDeleteNotify = 0
ScheduledDefrag disabled
No unexpected defrag.exe before start
```

Appendix A Step 2 - Run Standard Execution Block Through Step 12

Run Step 5-13 using:

```
OutputDir = C:\Temp\TrimTest-D-ManualRetrim
```

Important:

Do not stop the collectors after Step 13.

Appendix A Step 3 - Mark Before Manual ReTrim

Where:

Windows Server 2022 PowerShell – Run as Administrator

Run:

```
cd C:\Temp\TrimTest-Scripts

.\03-TrimLatencyTrace-kevin.ps1 -Action Mark -OutputDir $OutputDir -Message
"Before manual Optimize-Volume ReTrim"
```

Appendix A Step 4 - Run Manual Optimize-Volume / ReTrim

Where:

Windows Server 2022 PowerShell – Run as Administrator

Run:

```
Optimize-Volume -DriveLetter E -ReTrim -Verbose
```

Important:

Replace drive letter E with the actual target volume drive letter.
Run the command only after evidence collectors are already running.

Appendix A Step 5 - Mark After Manual ReTrim

Run:

```
cd C:\Temp\TrimTest-Scripts

.\03-TrimLatencyTrace-kevin.ps1 -Action Mark -OutputDir $OutputDir -Message
"After manual Optimize-Volume ReTrim"
```

Appendix A Step 6 - Observe 60+ Minutes and Complete Step 13-16

Run:

Observe for at least 60 minutes, then complete Step 14-17.

Interpretation:

If the environment remains stable after file deletion but degrades only after manual ReTrim,
that strengthens the case for an optimizer-driven reclaim trigger.

Appendix B - Tuning Note

The following items are intentionally excluded from the main execution plan and from Appendix A execution unless separately approved for a later tuning phase:

```
vol_tp_reclaim_limit
vol_tp_reclaim_idle_delay
```